2026 Vill. Mat A1 – 7. gyakorlat

Euler-szám, rendőrelv

1.

$$\lim_{n\to\infty}\left(1+\frac{\alpha}{n}\right)^n = e^{\alpha}$$

$$\lim_{n\to\infty}\left(\frac{n+4}{n+6}\right)^{n+2} =? \qquad \textbf{hf.:}\ \lim_{n\to\infty}\left(\frac{n-3}{n-7}\right)^{n+4} =?$$

2.

$$\exists \lim(a_n) = \lim(c_n) = A \quad \& \quad (\exists N \in \mathbf{N}\ \forall n > N a_n \le b_n \le c_n) \quad \Rightarrow \quad \exists \lim(b_n) = A$$

$$\lim_{n\to\infty} \sqrt[n]{p} = 1 \quad (p>0), \qquad \lim_{n\to\infty} \sqrt[n]{n} = 1$$

$$\textbf{a)}\ \lim_{n\to\infty} \sqrt[n]{n^4+5n^3+2} =?, \qquad \textbf{b)}\ \lim_{n\to\infty} \sqrt[n]{3^n-2n^4+1} =?$$

$$\textbf{gy.:}\ \lim_{n\to\infty} \sqrt[n]{\frac{4^n-3n^2}{n^2+n+1}} =?, \qquad \textbf{hf.:}\ \lim_{n\to\infty} \sqrt[n]{\frac{6^n+2n^7-n^5}{2^n-n+4}} =?$$

3.

$$\textbf{a)}\ \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n^2} =?, \qquad \textbf{b)}\ \lim_{n\to\infty}\left(1+\frac{1}{n^2}\right)^{n} =?$$

4.

$$\lim_{x\to 0}\frac{\ln(1+x)}{x} = 1, \qquad X^Y := e^{Y\cdot \ln X}$$

$$a)\ \lim_{x\to 0+}(1+2x)^{\frac{1}{x}}, \qquad b)\ \lim_{x\to 0+}(\cos x)^{1/x^2}$$

$$c)\ \lim_{x\to 0-}(1-3x)^{1/x^2}, \qquad d)\ \lim_{x\to 0-}(1+\sin(x^2))^{1/x}$$

$$e)\ \lim_{x\to 0+}(\cos x)^{1/\sin^2 x}, \qquad f)\ \lim_{x\to 0+}(1+\tan^2 x)^{-1/x}$$